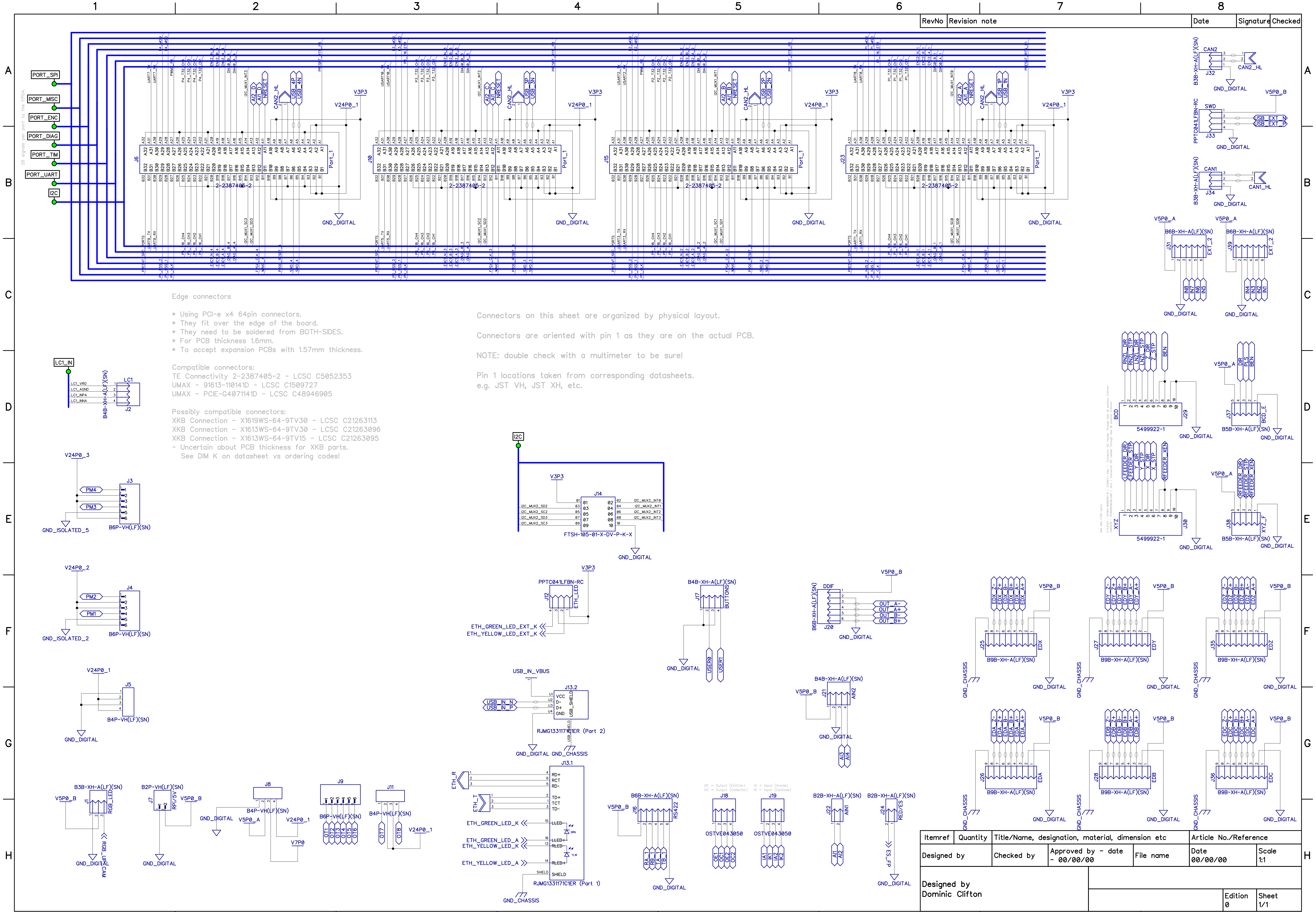

Edge connectors

* Using PCI-e x4 64pin connectors.
* They fit over the edge of the board.
* They need to be soldered from BOTH-SIDES.
* For PCB thickness 1.6mm.
* To accept expansion PCBs with 1.57mm thickness.

Compatible connectors:
TE Connectivity 2-2387405-2 - LCSC C5052353
UMAX - 91613-110141D - LCSC C1509727
UMAX - PCIE-G4071141D - LCSC C48946905

Possibly compatible connectors:
XKB Connection - X1619WS-64-9TV30 - LCSC C21263113
XKB Connection - X1613WS-64-9TV30 - LCSC C21263096
XKB Connection - X1613WS-64-9TV15 - LCSC C21263095
- Uncertain about PCB thickness for XKB parts.
  See DIM K on datasheet vs ordering codes!

Connectors on this sheet are organized by physical layout.

Connectors are oriented with pin 1 as they are on the actual PCB.

NOTE: double check with a multimeter to be sure!

Pin 1 locations taken from corresponding datasheets.
e.g. JST VH, JST XH, etc.

RevNo	Revision note	Date	Signature	Checked

Itemref	Quantity	Title/Name, designation, material, dimension etc	Article No./Reference	
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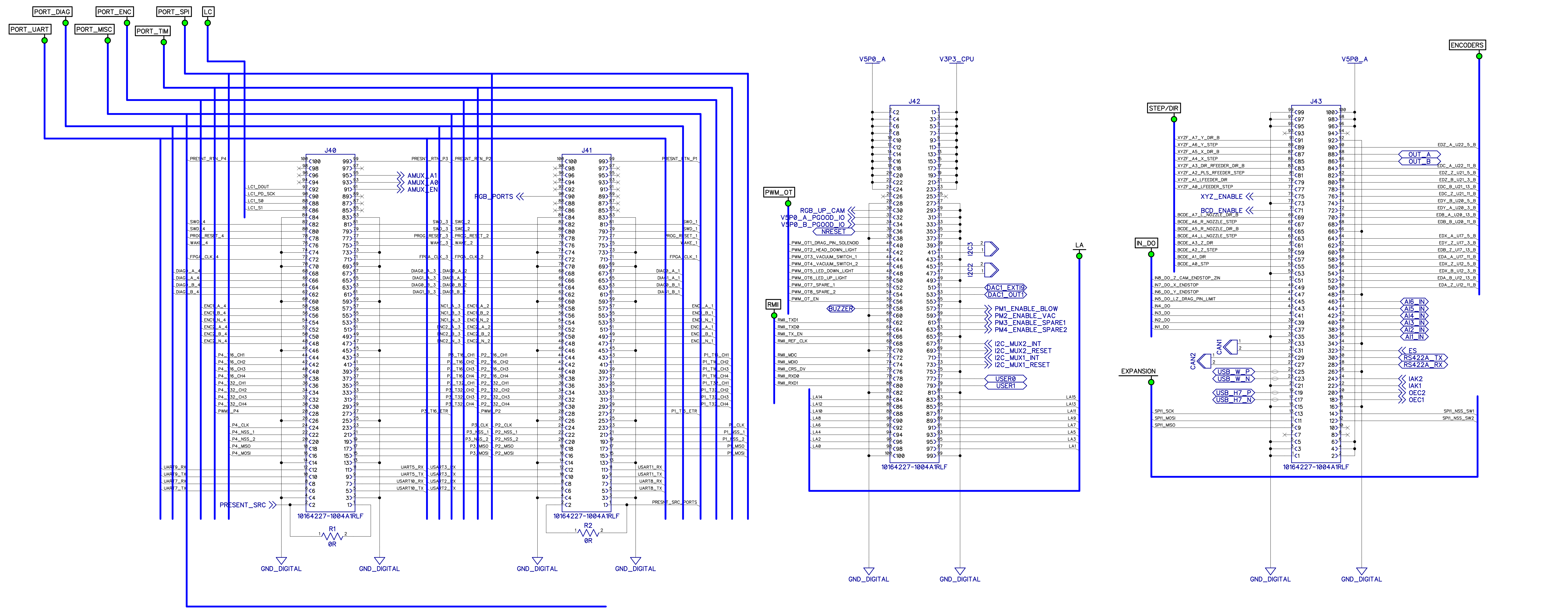

Card detection system

A pull-up to 3v3 is generated on the core board, signal is called PRESENT_SRC.
On the core board, this signal comes in on the left hand connector, pin 2.
The base board then routes this to pin 1 on the same connector.
The core board then routes pin 1 from the left connector to pin 2 on the right connector as PRESENT_THROUGH.
The base board this routes this under the right connector from pin 2 to pin 1.
The base board then routes this same signal from pin 1 on the right connector to each of the ports, as PRESENT_SRC_PORTS (aka PRESENT_BASE_RTN).
The port expansion cards then route the signal back to the FPGA on PRESENT_RTN_P1/2/3/4.

The FPGA then has 5 input pins.
One for pin 1 on right connector.
Then one for each PORT_RTN signal.

The FPGA can then detect correct seating of the core board.
The FPGA can then detect 4 different port expansion card correct seatings.

The pins chosen ensure each connector or expansion
card must be fully seated at both ends of the connector or slot.

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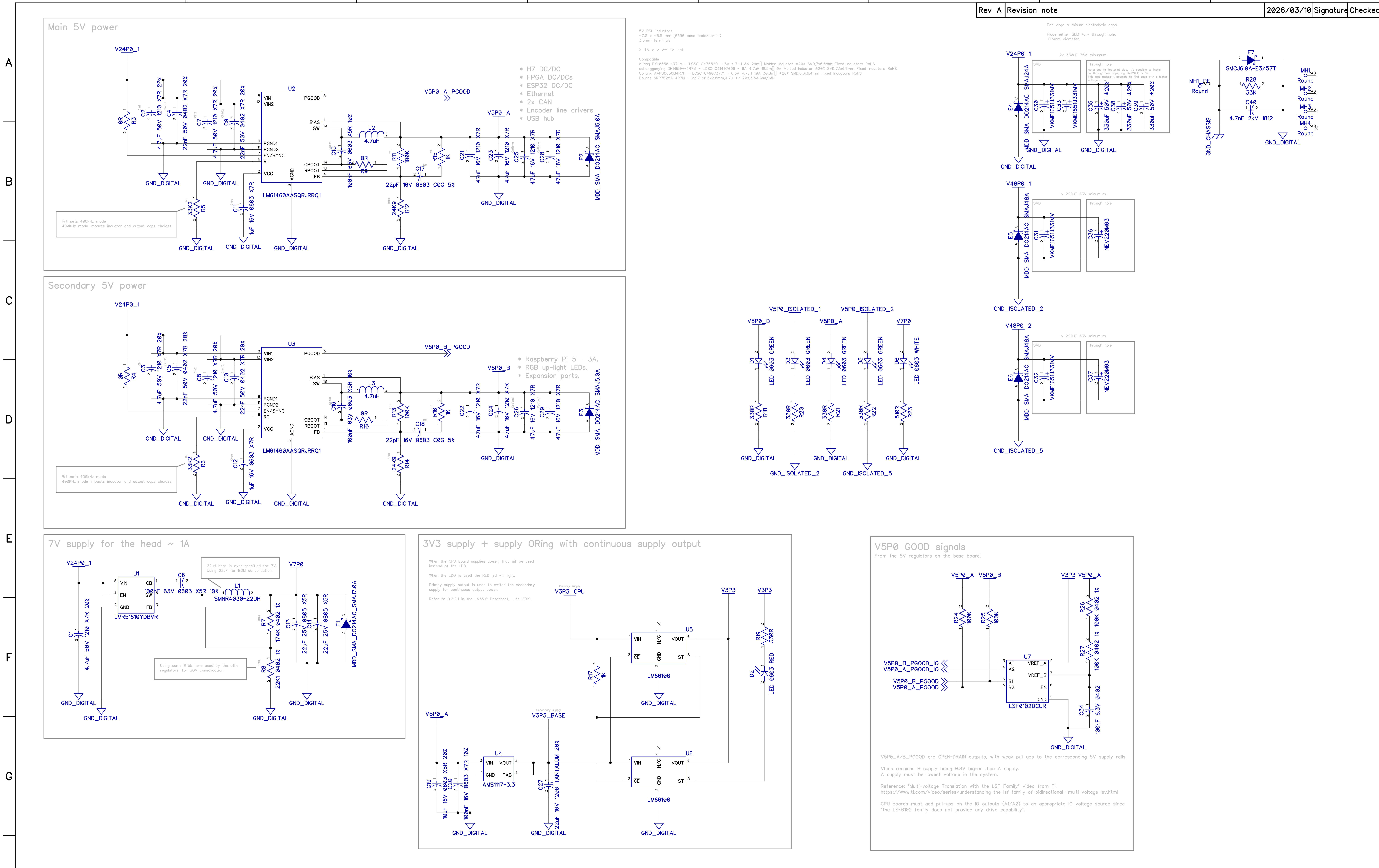

Itemref	Quantity	MakerPnP BASE		Article No./Reference	
Designed by	Checked by	Approved by - date - 00/00/00	File name	Date 2026/03/10	Scale 1:1
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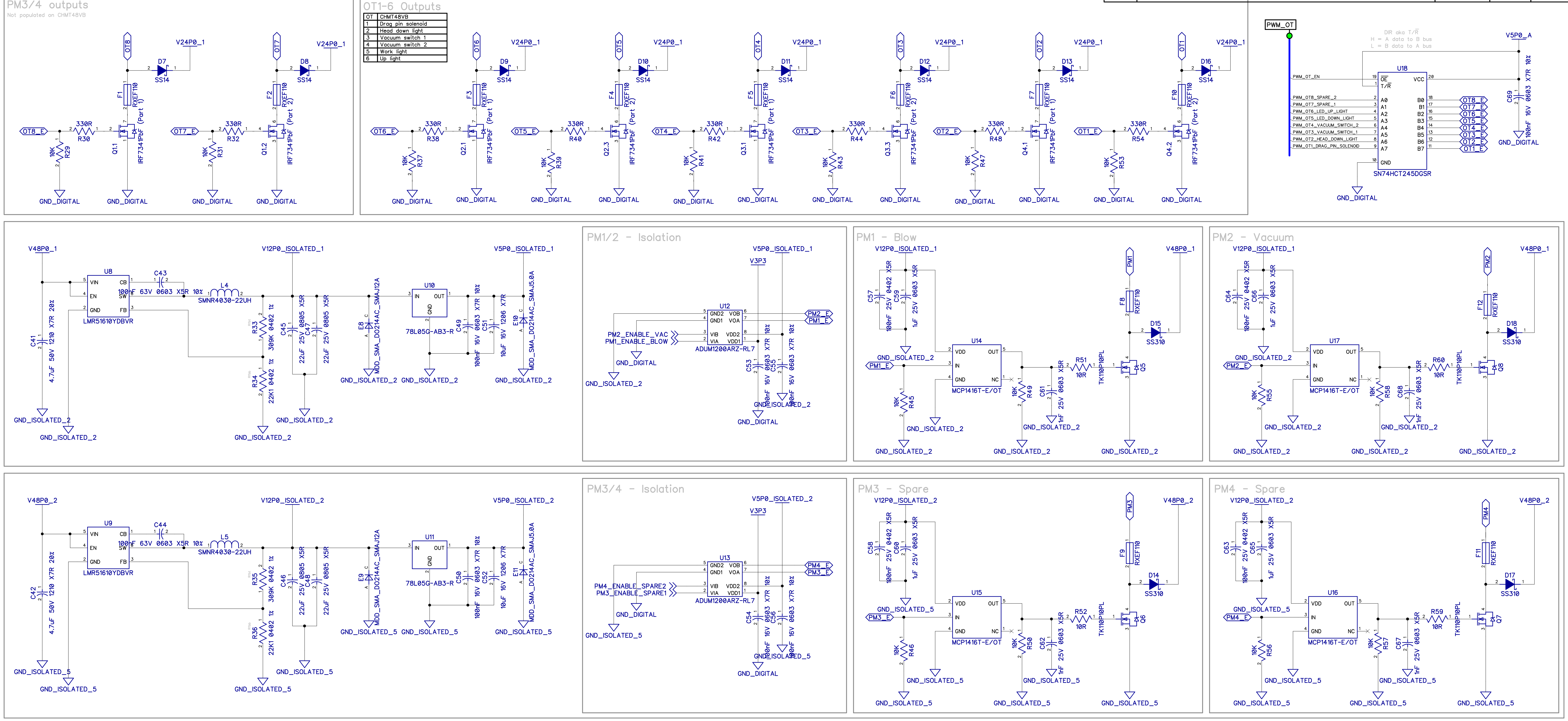
RevNo	Revision note	Date	Signature	Checked



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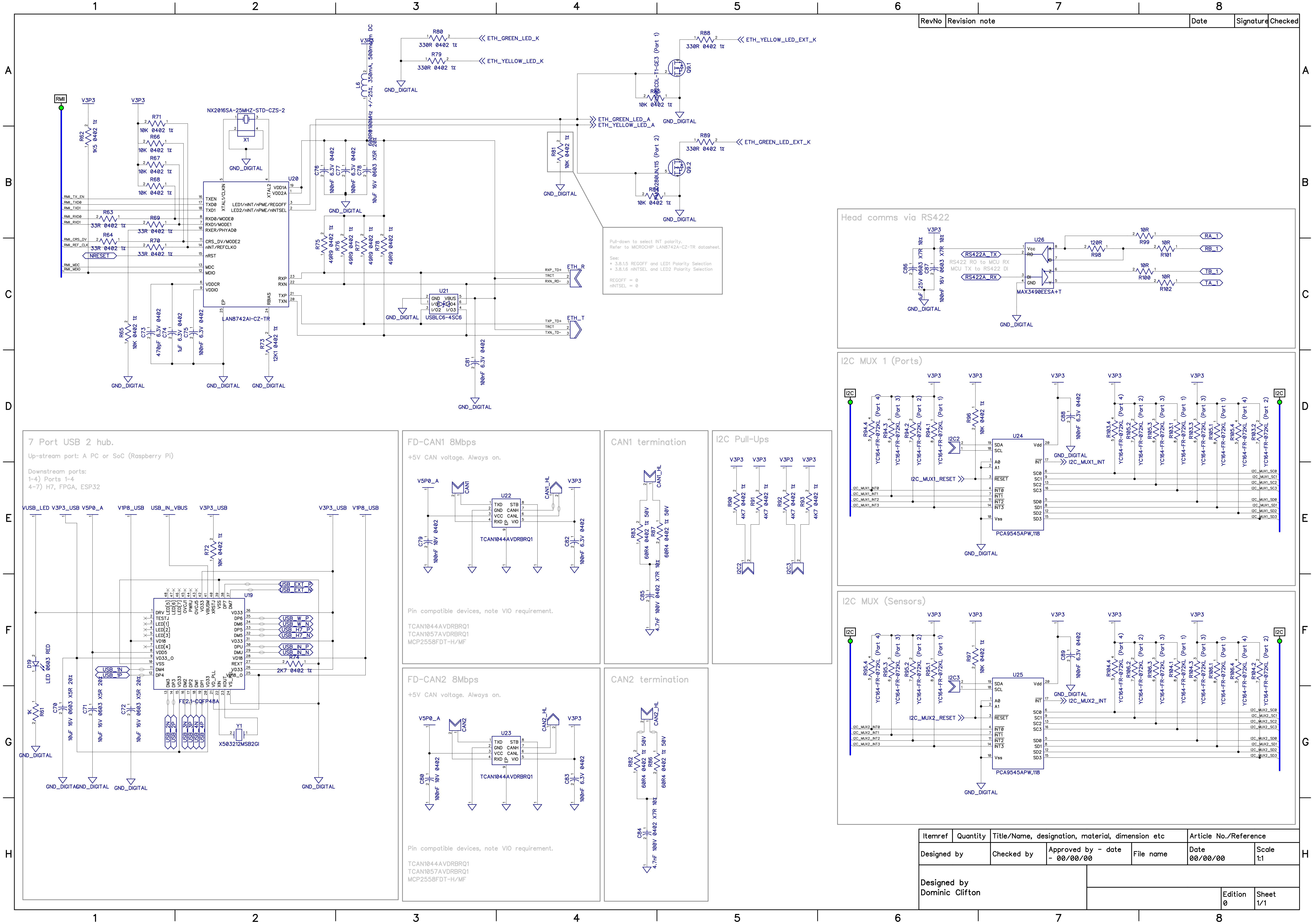
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NX2016SA-25MHZ-STD-CZS-2
LAN8742AI-CZ-TR
USBLC6-4SC6
RMII
NRESET
ETH_GREEN_LED_K
ETH_YELLOW_LED_K
ETH_GREEN_LED_A
ETH_YELLOW_LED_A
ETH_YELLOW_LED_EXT_K
ETH_GREEN_LED_EXT_K
ETH_R
ETH_T
Pull-down to select INT polarity.
Refer to MICROCHIP LAN8742A-CZ-TR datasheet.
See:
• 3.8.1.5 REGOFF and LED1 Polarity Selection
• 3.8.1.6 nINTSEL and LED2 Polarity Selection
REGOFF = 0
nINTSEL = 0
Head comms via RS422
RS422A_TX
RS422A_RX
RS422 RO to MCU RX
MCU TX to RS422 DI
MAX3490EESA+T
I2C MUX 1 (Ports)
I2C_MUX1_RESET
I2C_MUX1_INT
PCA9545APW,118
I2C MUX (Sensors)
I2C_MUX2_RESET
I2C_MUX2_INT
7 Port USB 2 hub.
Up-stream port: A PC or SoC (Raspberry Pi)
Downstream ports:
1-4) Ports 1-4
4-7) H7, FPGA, ESP32
FE2.1-CQFP48A
X50321ZMSB2GI
FD-CAN1 8Mbps
+5V CAN voltage. Always on.
TCAN1044AVDRBRQ1
Pin compatible devices, note VIO requirement.
TCAN1044AVDRBRQ1
TCAN1057AVDRBRQ1
MCP2558FDT-H/MF
FD-CAN2 8Mbps
+5V CAN voltage. Always on.
TCAN1044AVDRBRQ1
Pin compatible devices, note VIO requirement.
TCAN1044AVDRBRQ1
TCAN1057AVDRBRQ1
MCP2558FDT-H/MF
CAN1 termination
CAN2 termination
I2C Pull-Ups
Itemref
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1 2 3 4 5 6 7 8

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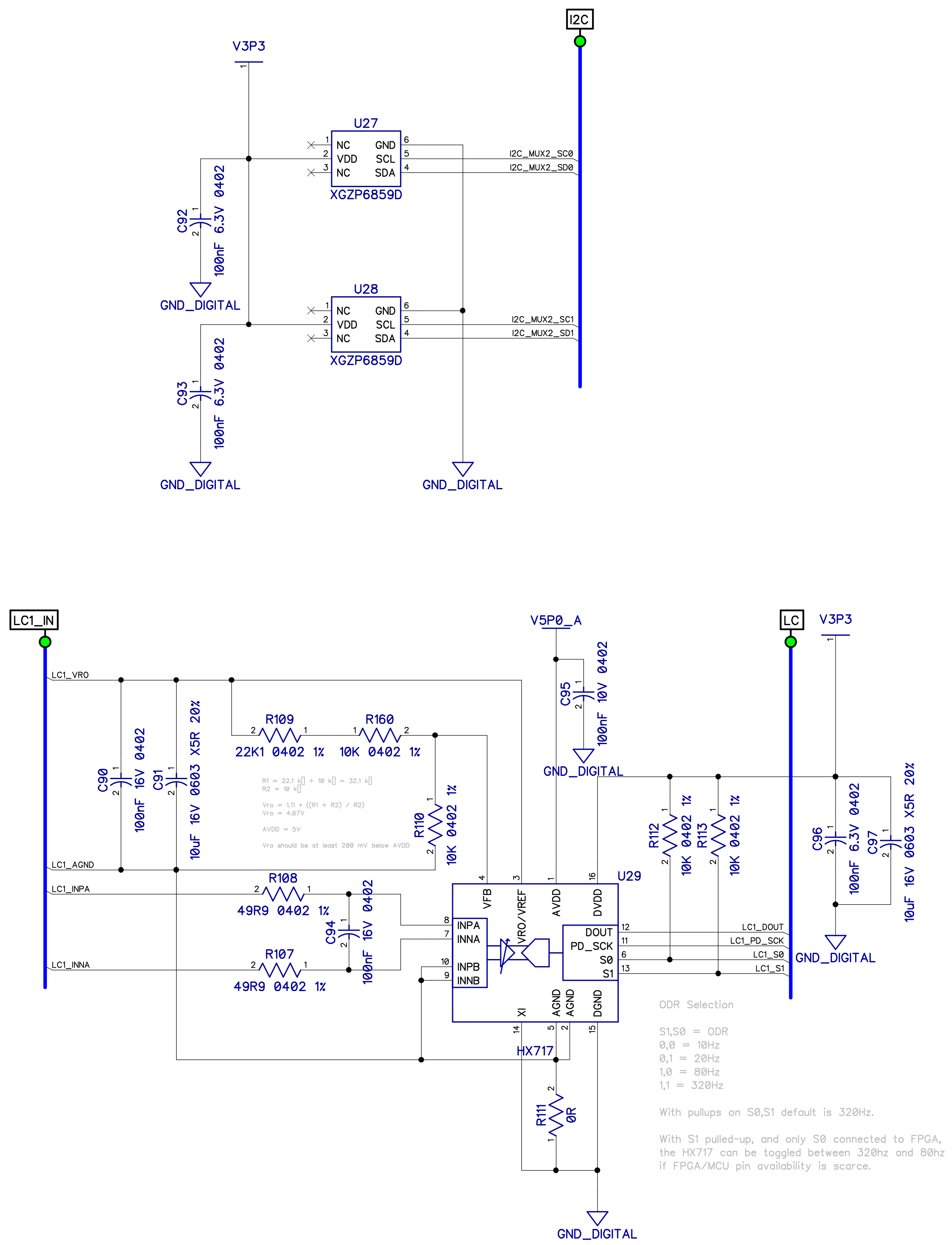

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Dominic Clifton

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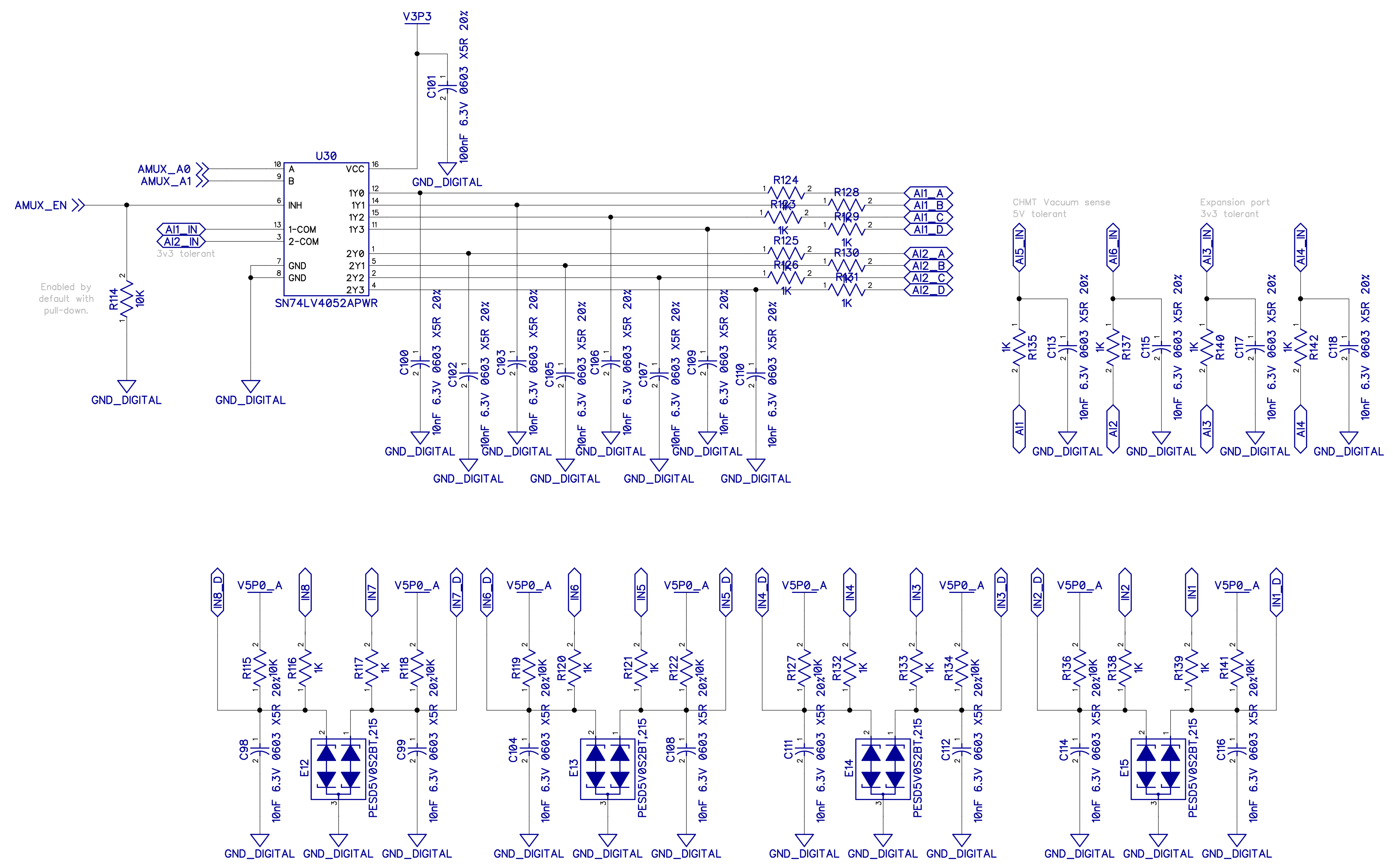

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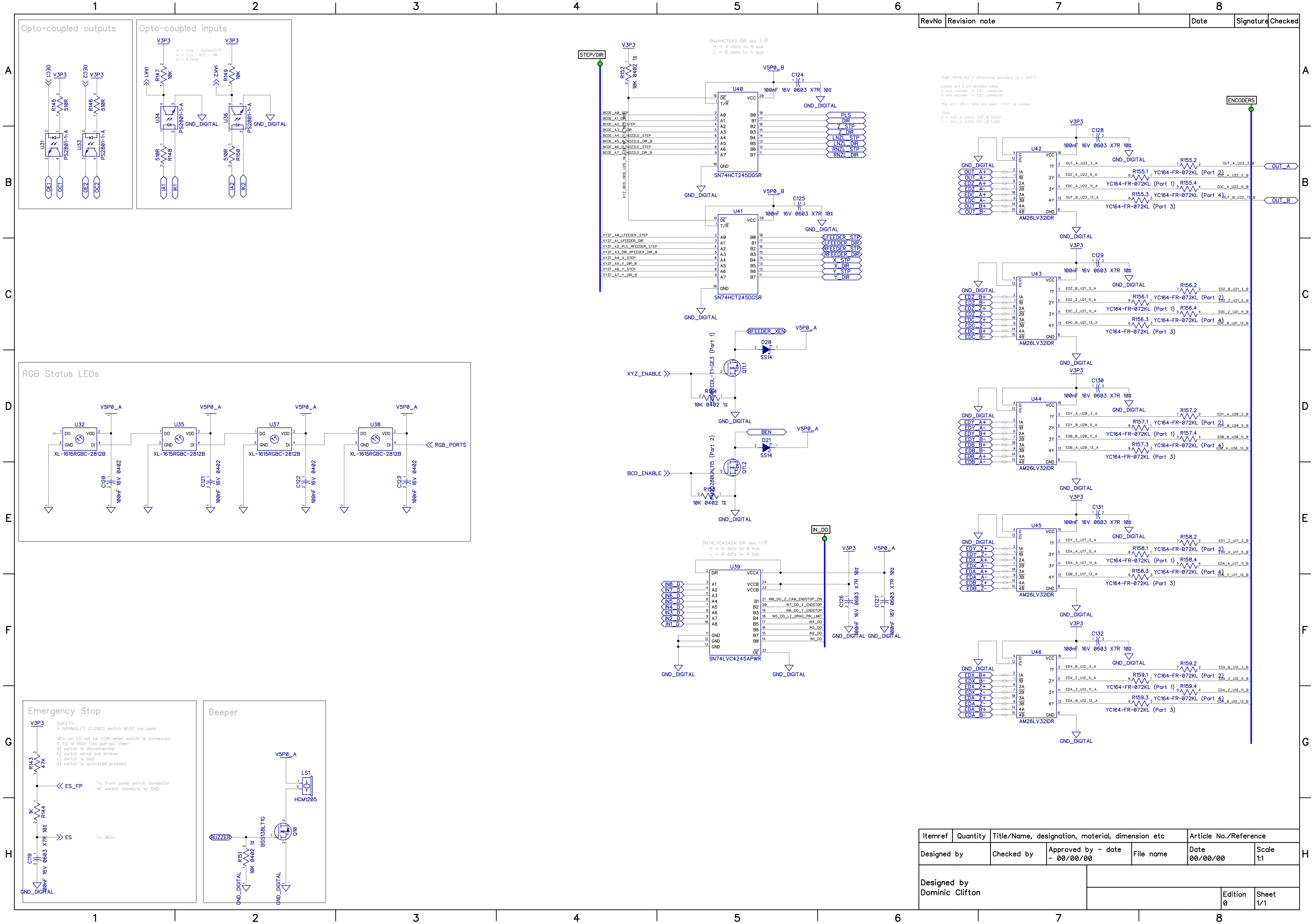

Opto-coupled outputs
Opto-coupled inputs
RGB Status LEDs
Emergency Stop
Beeper
SAFETY:
A NORMALLY CLOSED switch MUST be used.
To front panel switch connector
NC switch connects to GND.
To MCU.
STEP/DIR
ENCODERS
IN_DO
XYZ_ENABLE
BCD_ENABLE
RGB_PORTS
ES_FP
ES
BUZZER
SN74HCT245DGSR
SN74LVC4245APWR
AM26LV32IDR
XL-1615RGBC-2812B
PS2801-1-A
YC164-FR-072KL
HCM1205
BSS138LT1G
SS14
V3P3
V5P0_A
V5P0_B
GND_DIGITAL
OUT_A
OUT_B
RevNo
Revision note
Date
Signature
Checked
Itemref
Quantity
Title/Name, designation, material, dimension etc
Article No./Reference
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Checked by
Approved by - date
- 00/00/00
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